Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/01/2025
PI Name:	Garbino, Alejandro M.D., Ph.D.		
Project Title:	Effect of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario (PI: Garbino)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/14/2024	**End Date:**	12/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: Dr. Alejandro Garbino became Principal Investigator for this project in January, 2024 (Ed., 1/19/24). The previous PI, Dr. Derek Nusbaum, left the laboratory where these experiments were being conducted. Project is a continuation of "Effect of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario". Previous PI: Dr. Derek Nusbaum. June 2024 Report: PI notes that "Student Record" section is N/A (Ed., 7/1/24).		
COI Name (Institution):	Scully, Robert Ph.D. (KBR/NASA Johnson Space Center) Levine, Benjamin MD (University of Texas Southwestern Medical Center) Babb, Tony Ph.D. (University of Texas Southwestern Medical Center) Pawelczyk, James Ph.D. (Pennsylvania State University) Baughman, Alex M.S. (NASA Johnson Space Center) Siders, Brett (KBR/NASA Johnson Space Center) Keller, Nathan (KBR/NASA Johnson Space Center) Marshall-Goebel, Karina (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Task Objective/Description (updated by PI in June 2024) Carbon dioxide (CO2) may accumulate inside a space suit during extravehicular activities (EVA). The maximum allowable inspired CO2 during a 1 hour contingency walkback scenario is unknown. The purpose of this study is to characterize the limit of acceptable performance (cognitive and physical) decrements and symptom severity for mission operations when subjected to elevated inspired CO2 levels (up to 30 mmHg). Specific Aim 1 – To quantify operational physical and cognitive performance across a range of inspired CO2 levels during simulated suited contingency EVA operations in a ground-based, shirtsleeve, laboratory study using standardized research techniques. Hypothesis 1.1. – Physical performance (ambulation speed and distance) of simulated EVA tasks will be reduced in a non-linear fashion with increasing partial pressures of inhaled carbon dioxide. Hypothesis 1.2. – Cognitive performance (visual search, spatial memory, paired associate learning) during simulated EVA tasks will be reduced in a non-linear fashion with increasing partial pressures of inhaled carbon dioxide. Specific Aim 2 – To qualify and quantify subjective metrics, including symptoms and self-assessment of performance experienced across a range of inspired CO2 levels during simulated suited contingency EVA operations in a ground-based, shirtsleeve, laboratory study using standardized research techniques. Hypothesis 2.1. – Subjects will experience carbon dioxide-associated symptoms when exposed to elevated partial pressures of inhaled carbon dioxide. Hypothesis 2.2. - Non-linear increases in symptom severity will be observed with increasing partial pressures of inhaled carbon dioxide. Hypothesis 2.3. – Self-assessment of performance will be impaired by increasing partial pressures of inhaled carbon dioxide, despite decrements in objective performance. Methods Study Design - This study will use a repeated measures ground-based study design. A hybrid reality EVA simulation environment will provide a physically and cognitively realistic simulation of a 1 hour contingency walkback, incorporating a passive treadmill and a fully immersive virtual reality simulation of lunar EVA. This study will be performed in a shirtsleeve environment, along with a simulated spacesuit Hard Upper Torso, in a laboratory setting in NASA Johnson Space Center's Building 21 with subjects breathing compressed air mixed with the prescribed fraction of CO2. Subjects will be asked to walk on a treadmill at a self-selected speed within a defined range, bounded by a minimum of 3 km/hr and a maximum speed associated with a maximum allowable metabolic intensity. Cognitive performance will be quantified using validated cognitive tests, as well as measures of functional task performance embedded within the APACHE environment. Physical performance will be measured by evaluating the distance translated by subjects during a one-hour simulated contingency walkback on a treadmill. The severity of specific, CO2 related symptoms (headache, dizziness, nausea, numbness, tingling, shortness of breath, air hunger, fatigue, euphoria) will be quantified via survey prior to the testing session, at 10 minute intervals during the testing session, and immediately following the testing session, as well as every 10 minutes for one hour post exposure to document recovery. Fifteen subjects will be recruited to participate in this study; Each subject will be exposed to 7 different partial pressures of carbon dioxide (0, 5, 10, 15, 20, 25, 30 mmHg) for 1 hour at a time, each on a different day to provide sufficient time for washout of exposure. During the one hour of exposure, the subjects will breathe 21% oxygen compressed air (+CO2) via oronasal facemask. The order in which subjects are exposed to each of the different carbon dioxide partial pressures will utilize balanced randomization and the subjects will be blinded to the pressure of inspired carbon dioxide (PiCO2) levels. The range of CO2 levels to be used have been tolerated in previous studies.
Rationale for HRP Directed Research:	This project is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	Updated by PI in June 2024 The principal benefits of this study are expected to be in the area of operational space medicine, where these data will help inform future planetary EVA suit design and may drive EVA operational constraints. However, as this study deals with the cognitive and physical performance aspects of elevated CO2 exposure, it may have broader implications for clinical patients who are also exposed to elevated CO2, such as patients with chronic obstructive lung disease or patients with obesity hypoventilation syndrome.
Task Progress:	The study developed a novel breathing gas delivery system that allows a customized delivery and monitoring of a wide range of CO2 levels. Using that system and existing lab tools (HS3 suit simulator, VR system, treadmill with overhead suspension), the team successfully carried out over 120 HITL exposures, measuring various physiologic parameters for CO2 exposures spanning 0-30mmHg piCO2 with 15 subjects. A new "CO2 contingency standard" for space suit operations was created and released on Rev E of NASA-STD-3001 v2.
Bibliography Type:	Description: (Last Updated: 04/28/2026)
Abstracts for Journals and Proceedings	Keller N, Nusbaum DM, Scully R, Levine B, Babb T, Pawelczyk J, Baughman A, Basner M, Abercromby A, Garbino A. "Effect of carbon dioxide exposure on physical and cognitive performance in a simulated spaceflight contingency scenario." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts.2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Keller N, Scully R, Levine B, Babb T, Pawelczyk J, Baughman A, Basner M, Marshall-Goebel K, Garbino A. "Time course of effects of carbon dioxide exposure on physical and cognitive performance in a simulated surface extravehicular activity contingency scenario." 95th Annual Scientific Meeting of the Aerospace Medical Association (AsMA), Atlanta, GA, June 1-6, 2025. Abstracts. 95th Annual Scientific Meeting of the Aerospace Medical Association (AsMA), Atlanta, GA, June 1-6, 2025. , Jun-2025

Articles in Peer-reviewed Journals	Keller N, Thoolen S, Siders B, Estep P, Scully R, Levine BD, Babb T, Pawelczyk J, Baughman A, Basner M, Young M, Chappell L, Norcross J, Marshall-Goebel K, Garbino A. "Effects of carbon dioxide on physical and cognitive performance in a simulated spacesuit contingency scenario." npj Microgravity. 2026 Apr 21. https://doi.org/10.1038/s41526-026-00589-x ; PMID: 42014415 , Apr-2026